

RESEARCH PROFILE

Biomedical Engineer, Magna Cum Laude, from Universidad de los Andes, currently pursuing an M.Sc. in Biomedical Engineering with a focus on Biomedical Computer Vision under the supervision of Prof. Pablo Arbeláez. I have 2 years of research experience in artificial intelligence for biomedical applications, with work spanning multimodal learning, foundation models and representation learning for medical imaging and spatial transcriptomics. I have contributed to work accepted or published at MICCAI 2026 and an ICCV 2025 Workshop, and led the development of a foundation-model-based solution for the EchoRisk-MICCAI Cardio-Oncology Challenge. In parallel, I am a final-semester Instrumental Music student with a Piano emphasis at Universidad Nacional de Colombia and a classically trained pianist. This dual academic path has strengthened my discipline, organization, adaptability, analytical thinking, and creativity.

EDUCATION

M.Sc. in Biomedical Engineering <i>Universidad de los Andes Current GPA: 4.91/5.00</i>	Jan. 2025 – Present <i>Bogotá D.C., Colombia</i>
• Research Focus: Biomedical Computer Vision, Spatial Transcriptomics, Multimodal Learning, and Foundation Models. • Relevant Coursework: Machine Learning, Fundamentals of Research in Artificial Intelligence, Computer Vision I: Visual Interpretation, Computer Vision II: Beyond Pixels.	
B.Sc. in Biomedical Engineering — Magna Cum Laude <i>Universidad de los Andes GPA: 4.80/5.00</i>	Jan. 2021 – Apr. 2025 <i>Bogotá D.C., Colombia</i>
• Top 1% historical GPA distribution among Faculty of Engineering graduates over the previous five years. • Relevant Coursework: Scientific Programming, Probability and Statistics, Image Analysis and Processing, Experimental Design and Biostatistics, Biomedical Modeling and Simulation, Molecular Biology. • Undergraduate Thesis: <i>Noap: An Innovative Service for the Treatment of Obstructive Sleep Apnea</i>.	
Bachelor of Music (B.M.) – Piano Performance <i>Universidad Nacional de Colombia GPA: 4.60/5.00</i>	Aug. 2021 – Present <i>Bogotá D.C., Colombia</i>
• Final-semester student; ranked among the Top 15 students by academic performance for four semesters.	

RESEARCH EXPERIENCE

Graduate Research Assistant <i>Biomedical Computer Vision Group (BCV), Universidad de los Andes</i>	Aug. 2024 – Present <i>Bogotá D.C., Colombia</i>
• M.Sc. Research — Spatial Gene-Expression Prediction: Developing multimodal and representation-learning methods for spatial gene-expression prediction from histopathology, integrating histological, gene-expression, and spatial representations across heterogeneous datasets as part of my M.Sc. thesis. • EchoRisk-MICCAI — Echocardiography Foundation Models: Led a four-researcher team developing a foundation-model-based solution for cardiac risk prediction from echocardiography; adapted Echo-Vision-FM and EchoCLIP through fine-tuning, multi-view modeling, ensembling, and test-time augmentation. Served as first and corresponding author and led experimental integration, results analysis, and manuscript preparation. • HyCLOST / SpaRED — Spatial Transcriptomics: Integrated recent state-of-the-art methods into the SpaRED benchmark, enabling standardized and reproducible evaluation across 26 spatial transcriptomics datasets; contributed to experimental design, benchmarking, results analysis, manuscript preparation, reviewer response, and technical restructuring. • CARDIUM — Multimodal Prenatal Diagnosis: Contributed to the construction of a multimodal prenatal diagnostic dataset by extracting and structuring 26 maternal clinical variables; designed and optimized the tabular encoder and contributed to multimodal fusion using cross-attention between clinical and visual representations.	
Nanobiomaterials, Cellular Engineering, and Bioprinting <i>Universidad de los Andes</i>	2024 <i>Bogotá D.C., Colombia</i>
• Conducted microfluidic fabrication and characterization experiments and performed ImageJ-based quantitative analysis of laser-ablation surface roughness and nanobioconjugate morphology.	

SELECTED PUBLICATIONS

Task-Specific Adaptation of Echocardiography Foundation Models for the EchoRisk-MICCAI Cardio-Oncology Challenge [Paper]	2026
H. Ceballos, N. Andrade, L. Mora-Vaca, J.D. Sarmiento Avila, D. Vega, E.E. Romero Castro, P. Arbeláez <i>Accepted at the EchoRisk-MICCAI 2026 Satellite Event, jointly with the STACOM Workshop.</i>	
Hyperbolic Contrastive Learning with Entailment for Spatial Transcriptomics [Paper]	2026
D. Vega, P. Cárdenas, H. Ceballos, L. Manrique, P. Arbeláez <i>Accepted at the Medical Image Computing and Computer-Assisted Intervention Conference (MICCAI).</i>	

CARDIUM: Congenital Anomaly Recognition with Diagnostic Images and Unified Medical Records [Paper]	2025
D. Vega, H.V. Ceballos , J.S. Vera, S. Rodriguez, A. Perez, A. Castillo, et al. <i>Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops</i> , pp. 1193–1202.	
Restoring skeletal remodeling in antiresorptive-treated patients: clinical outcomes of teriparatide in medication-related osteonecrosis of the jaw [Paper]	2026
A.K. Sarmiento Liévano, H.V. Ceballos Sarmiento , et al. <i>Osteoporosis International</i> .	

HONORS & AWARDS

Magna Cum Laude — Biomedical Engineering	2025
<i>Universidad de los Andes</i> Bogotá D.C., Colombia	
Graduation distinction corresponding to the top 1% historical GPA distribution among Faculty of Engineering graduates over the previous five years.	
Top 10 Academic Performance — Biomedical Engineering	Four consecutive semesters
<i>Universidad de los Andes</i> Bogotá D.C., Colombia	
Top 15 Academic Performance — Instrumental Music	Four semesters
<i>Universidad Nacional de Colombia</i> Bogotá D.C., Colombia	
Selected Representative — Undergraduate Thesis Abstract Series	2026
<i>Universidad Nacional de Colombia</i> Bogotá D.C., Colombia	
Selected as the representative undergraduate thesis from the Bogotá campus for inclusion in Universidad Nacional de Colombia's <i>Abstract de Trabajos de Grado</i> series.	

PROFESSIONAL & TEACHING EXPERIENCE

Junior Researcher	Jul. 2026 – Present
<i>Ministry of Science, Technology and Innovation — Orquídeas Program</i> Universidad de los Andes	
Curate jaguar camera-trap imagery, retrain species-classification models in Sparrow Studio/SPARROW, and support integration of Microsoft-developed tools for Magdalena Medio data.	
Graduate Teaching Assistant — Computer Vision I & II	Jan. 2026 – May 2026
<i>Master's Program in Artificial Intelligence (MAlA), Universidad de los Andes</i> Bogotá D.C., Colombia	
- Led laboratories, tutorials, and weekly mentoring sessions covering CNNs, Vision Transformers, object detection and segmentation, 3D reconstruction, video analysis, and generative models. - Supported student assessment, technical feedback, and problem-solving sessions across both graduate-level computer vision courses.	

SELECTED CONFERENCES & WORKSHOPS

MICCAI 2026	Strasbourg, France — Scheduled attendee and poster presenter for two accepted works: a main-conference paper on spatial transcriptomics and the EchoRisk-MICCAI Cardio-Oncology Challenge / STACOM.
ICCV 2025 — CVAMD Workshop	Co-authored <i>CARDIUM: Congenital Anomaly Recognition with Diagnostic Images and Unified Medical Records</i> , which was presented as a poster at the workshop.

TECHNICAL SKILLS

Programming Languages	Python, Bash.
Scientific Computing	NumPy, SciPy, Pandas, scikit-image.
ML Frameworks	PyTorch, scikit-learn, Hugging Face Transformers.
Tools & Infrastructure	Git, Docker, Linux, Weights & Biases.
Languages	Spanish (Native), English (C1 — TOEFL iBT 105/120).

ADDITIONAL TRAINING & ARTISTIC BACKGROUND

Microcredential: Research and Consulting Project Formulation and Structuring, CIDER — Universidad de los Andes, 2025.

Classical Piano

- Final-semester Instrumental Music student with a Piano emphasis at Universidad Nacional de Colombia.
- Winner**, Contra Corriente University Festival, 2023.
- Honorable Mention**, IX Pianissimo Festival and Competition, 2022.